

Jerry Xizhao Fu

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Student building intelligent systems across robotics, medicine, and AI.

Education

2025–(2027) Collège Jean-de-Brébeuf – IBDP + DEC Pure and Applied Sciences (pre-uni)

2020–2025 Collège Regina Assumpta – Secondary I–V, Natural Sciences and Robotics

Research Experience

2025–present **CHUM Research Center – Research Assistant**

- Processed lung ultrasound elastography sequences, segmented the pleural line, ran strain analysis in MATLAB, and debugged processing scripts. Contributed to normative regional pleural strain values and their correlation with tidal volume.
- Calibrated spirometry equipment and assisted during volunteer sessions. Tuned ultrasound acquisition parameters for a follow-on biplanar-probe study.
- Observed in the Operating Room and Intensive Care Unit at the CHUM: cardiac surgery with cardiopulmonary bypass, intraoperative transesophageal echocardiography, epidural placement, intubation, continuous renal replacement therapy, and more.

Publications/Abstracts

- Sep 2025 · CHUM Research Center *Relation entre la déformation pleurale régionale mesurée par élastographie pulmonaire et le volume courant chez des volontaires sains*. S. Chraïbi, K. El Berhoumi, A. Danatou, J. Fu, P. Vianna, M.H. Roy Cardinal, G. Cloutier, M. Girard (presenting author). Annual Congress of the French Society of Anesthesia and Critical Care (SFAR), Paris, France.

Work Experience

2023–present **DODO Wheels / PublicoTires – Software Developer**

- Collaborating with a small team to build a centralized operations platform unifying warehouse stock, inventory, invoicing, customer, and product management across the company (in progress).
- Built an OAuth 2.0 integration layer between the company website and QuickBooks Online, enabling programmatic management of invoices, customers, and inventory.
- Automated the migration of 3,000+ products and images to the new system via a web scraper.
- 2024 Developed the Doublestar Tire Canada website — doublestartyre.ca

2022–present **FIRST Robotics Competition – Mentor & former Student of Team 3990**

- Created coursework and taught programming, computer vision and AI/ML to junior team members.
- Developed computer vision object tracking and autonomous behaviors on the robot.
- Multiple consecutive Regional Winners; 2024 World Championship Impact Award finalist.

Volunteer Experience

- 2025 Developed a score-tracking application for our in-house off-season robotics competitions.
- 2024 Volunteered at our in-house FIRST event and presented our robotics program.
- 2025 *Anges Gardiens* volunteer; peer guidance to new students at Collège Regina Assumpta.
- 2025 Volunteer photographer for dance show at Collège Regina Assumpta in intricate lighting.

Extracurriculars/Notable Projects

- **2026 WRCC Brain–Computer Interface Competition** – Contributed to an EEG motor-imagery classifier using a Riemannian geometry pipeline on four channels, averaging 77% accuracy across four binary tasks. Qualified for the finals.
- **2025–2026 ORCA – Operational Intelligence Platform** – Designed a distributed pipeline ingesting telemetry, sensor, and radio data through a streaming entity graph to produce real-time, cited assessments. Prepared a TRL-3 proposal for the Canadian DND/CAF IDEaS program.
- **2025–2026 Medical Inference via Vector Embedding** – Personal research project developing an AI system using vector embeddings to model symptom–condition relationships for disease prediction.
- **HackQC 2024** – Developed a DiscordJS chatbot using OpenAI LLMs and LangChain to query Quebec Open Data, providing users with fast, searchable information.
- **2024 & 2025 *Secondaire en spectacle*** Performed piano-comedy skits with a friend, 400+ audience. Local winner.

Programming Skills

- **Languages:** TypeScript/JS, Python, Java, Kotlin, Swift, MATLAB
- **Frameworks & Libraries:** React, PyTorch, OpenCV, WPILib, Kotlin Multiplatform, SwiftUI
- **Infrastructure:** Docker, Cloudflare, GCP, OAuth 2.0

Other Skills

- Hands-on experience with tools, electronics, and machines
- Basic CAD (Onshape, Autodesk Inventor)
- Photography and Adobe Lightroom

Certifications

- Good Clinical Practice (CITI), Éthique de la recherche (MSSS), Responsible Conduct of Research
- Canadian Red Cross: Standard First Aid & CPR/AED Level C, Psychological First Aid